

Audio Emotion Recognition Using Wav2Vec2 and the LAION Emilia Dataset

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Abstract—This paper presents a lightweight approach for continuous emotion recognition from speech using the facebook/wav2vec2-base-960h model. The objective is to predict dimensional emotional attributes including Valence, Arousal, Dominance, and Humor directly from raw audio. Leveraging a subset of the LAION Emilia dataset, a pretrained Wav2Vec2 encoder is used as a frozen feature extractor, while a small GRU-based regression head with attention pooling is trained on top of the encoder outputs. Labels are normalized to a common $[0, 1]$ range during training and mapped back to their original scales at inference. Model performance is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and Concordance Correlation Coefficient (CCC) for each attribute. The study demonstrates that training only a lightweight head on top of frozen self-supervised speech representations can provide effective emotion regression while greatly reducing computational cost.

Index Terms—audio emotion recognition, wav2vec2, self-supervised learning, speech processing, regression, affective computing

I. INTRODUCTION

Speech carries rich emotional information through intonation, pitch, and rhythm that is not evident from text alone. Automatically recognizing such emotions from audio enables more natural and context-aware human–computer interaction. This work focuses on audio-based emotion recognition using a dimensional representation of affect, predicting continuous values for attributes such as valence (positivity vs. negativity), arousal (calm vs. excited), dominance (submissive vs. dominant), and a seriousness–humor dimension.

The proposed system is built on wav2vec2-base-960h, a self-supervised speech model pretrained on a large amount of English speech. Rather than fine-tuning all model parameters, we freeze the entire Wav2Vec2 encoder and train only a task-specific regression head on top of its hidden representations. This strategy makes training significantly more efficient and hardware-friendly while still benefiting from the rich representations learned during pretraining.

II. MOTIVATION

Many real-world applications—such as virtual assistants, call center analytics, meeting summaries, audio searches, and mental health support—can benefit from systems that recognize how a speaker feels, not just what they say. However,

training large transformer-based speech models end-to-end for emotion recognition is often computationally expensive and challenging to deploy on resource-constrained hardware.

By reusing the pretrained wav2vec2-base-960h encoder as a frozen feature extractor, this project aims to achieve a good trade-off between performance, robustness, and efficiency. Training only a relatively small GRU-based regression head greatly reduces the memory footprint, speeds up experimentation, and lowers the risk of overfitting when using limited labeled emotional data.

III. RELATED WORK

Speech emotion recognition (SER) has been an active area of research for several decades, evolving from traditional signal-processing approaches to modern deep learning and self-supervised representation learning frameworks. Existing work can broadly be categorized into handcrafted feature-based methods, supervised deep learning approaches, and recent self-supervised and pretrained speech representation models.

A. Traditional Feature-Based Emotion Recognition

Early SER systems relied heavily on handcrafted acoustic features such as Mel-Frequency Cepstral Coefficients (MFCCs), pitch, energy, formants, and speaking rate. These features were typically combined with classical machine learning classifiers or regressors, including Support Vector Machines (SVMs), Gaussian Mixture Models (GMMs), and Hidden Markov Models (HMMs). Benchmark efforts such as the INTERSPEECH 2009 Emotion Challenge [1] standardized datasets and evaluation protocols, establishing strong baselines for categorical and dimensional emotion recognition. While effective to an extent, these methods required extensive feature engineering and often struggled to generalize across speakers and recording conditions.

B. Deep Learning Approaches for Speech Emotion Recognition

With the rise of deep learning, researchers began replacing handcrafted features with learned representations. Convolutional Neural Networks (CNNs) were applied to spectrograms and log-Mel features to capture local time–frequency patterns,

while Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks were used to model temporal dependencies in speech signals. Satt *et al.* [2] demonstrated that deep neural architectures could outperform traditional approaches by learning discriminative emotional representations directly from acoustic inputs. However, these supervised models typically required large labeled datasets and were sensitive to data scarcity, which is a common issue in emotion annotation.

C. Dimensional Emotion Modeling

Beyond categorical emotion labels, many studies adopt a dimensional representation of emotion, most commonly Valence, Arousal, and Dominance (VAD). Dimensional modeling provides a more fine-grained and continuous representation of emotional state, which is particularly suitable for real-world applications. However, regression-based SER is more challenging than classification due to label noise, subjectivity in annotations, and the need to capture subtle emotional variations over time. Evaluation metrics such as Mean Squared Error (MSE), Mean Absolute Error (MAE), and Concordance Correlation Coefficient (CCC) have therefore become standard in this setting.

D. Self-Supervised Speech Representation Learning

Recent advances in self-supervised learning have significantly changed the SER landscape. Models such as wav2vec 2.0 [3] learn high-quality speech representations directly from raw waveforms by leveraging large amounts of unlabeled data. These representations capture phonetic, prosodic, and temporal characteristics of speech and have been shown to transfer effectively to downstream tasks, including emotion recognition.

Pepino *et al.* [4] demonstrated that wav2vec 2.0 embeddings significantly improve SER performance when fine-tuned on emotion datasets. Similarly, multilingual extensions such as XLS-R [5] showed strong cross-lingual generalization, enabling emotion recognition in low-resource languages. While these approaches often rely on full or partial fine-tuning of the pretrained encoder, such strategies can be computationally expensive and prone to overfitting when labeled data is limited.

E. Efficient and Parameter-Efficient Fine-Tuning

To address computational constraints, recent work has explored parameter-efficient fine-tuning strategies, including freezing large portions of pretrained models and training lightweight task-specific heads. These approaches reduce memory usage and training time while still benefiting from rich pretrained representations. In the context of SER, freezing the speech encoder and training only a regression head has been shown to be effective, particularly when combined with temporal pooling or attention mechanisms.

The approach adopted in this work follows this line of research by using a frozen Wav2Vec2-base encoder and a GRU-based regression head. This design balances performance and efficiency, making it suitable for resource-constrained

environments while still achieving competitive results on continuous emotion prediction tasks.

F. Summary

In summary, prior work demonstrates a progression from handcrafted acoustic features to deep learning and self-supervised speech representations for emotion recognition. While full fine-tuning of large pretrained models yields strong performance, it comes at a high computational cost. This work builds upon recent advances in self-supervised learning and efficient fine-tuning by investigating a lightweight regression-based approach for continuous emotion recognition using pretrained Wav2Vec2 representations.

IV. METHODOLOGY

This section describes the complete pipeline used in the project, including data preprocessing, feature extraction, regression head design, training procedure, and prediction of emotion attributes.

A. Dataset and Label Normalization

The project operates on audio files paired with JSON annotation files derived from the LAION Emilia emotion dataset. For each utterance, the following four continuous attributes are used as targets:

- Valence_best (range: -3 to 3),
- Arousal_best (range: 0 to 4),
- Submissive_vs._Dominant_best (range: -3 to 3),
- Serious_vs._Humorous_best (range: 0 to 4).

To facilitate stable training, each attribute is normalized to $[0, 1]$ using its known minimum and maximum:

$$x_{\text{norm}} = \frac{x - \min}{\max - \min}. \quad (1)$$

These normalized values are used as regression targets. At inference time, model outputs in $[0, 1]$ are denormalized back to the original attribute ranges using the inverse transformation so that predictions are directly interpretable.

B. Audio Preprocessing

Audio is loaded from .mp3 files using torchaudio. For each file:

- the original sampling rate is inferred using `torchaudio.load`;
- only the first 12 seconds of audio are loaded to limit computation;
- multi-channel audio is converted to mono by averaging channels;
- if the original sampling rate differs from the target 16 kHz, a cached `torchaudio.transforms.Resample` instance is used to resample the waveform;
- the waveform is truncated or zero-padded to a fixed length of $16,000 \times 12$ samples so all inputs have identical duration;

- the audio is peak-normalized by dividing by its maximum absolute value.

This preprocessing ensures consistent, normalized inputs suitable for Wav2Vec2 and allows the use of `padding="do_not_pad"` with the feature extractor.

C. Lazy Feature Extraction with Wav2Vec2-Base-960h

The model uses the pretrained `facebook/wav2vec2-base-960h` checkpoint. The implementation loads:

- `AutoFeatureExtractor` for waveform-to-input conversion, and
- `Wav2Vec2Model` to obtain contextualized hidden states.

A custom `LazyAudioData` class is used to lazy-load and cache data per file:

- on first access, it loads and preprocesses the waveform;
- the waveform is passed through the frozen Wav2Vec2 encoder to obtain `last_hidden_state` of shape (T', d_{model}) ;
- this tensor is cached as encoded features, and the waveform is discarded to save memory;
- the tensor is also saved so it can be accessed in future program runs without processing raw audio;
- subsequent accesses reuse the cached encoder output, avoiding recomputation.

A list of `LazyAudioData` objects is created by pairing `.mp3` files with their corresponding JSON annotation files. The dataset is then randomly shuffled and split into training and validation sets, with 10% of items used for validation and 90% used for training.

D. Regression Head Architecture

On top of the frozen encoder output, the project uses a GRU-based temporal head with attention pooling. The head:

- takes the sequence of encoder hidden states of shape $(B, T', d_{\text{model}})$ as input;
- processes it using a bidirectional GRU with hidden size 128, producing output of size $(B, T', 2 \times 128)$;
- applies an attention mechanism (two linear layers with `tanh` and a scalar score) to compute a weight for each time step;
- normalizes these weights with a softmax over time;
- forms a weighted sum of the GRU outputs to obtain a single pooled representation per utterance;
- maps this representation to a 4-D output vector corresponding to the four normalized emotion attributes using a final linear layer.

Only the parameters of this regression head are trainable; all Wav2Vec2 encoder parameters are frozen (`requires_grad = False`), which significantly reduces training cost.

E. Training Procedure

Training is carried out using:

- loss function: `nn.MSELoss` on the normalized targets,
- optimizer: AdamW applied only to the head parameters,
- batch size: 1 (helps with stability and avoids padding issues across examples),
- up to 50 epochs of training,
- learning rate: 1×10^{-3} ,
- weight decay: several trials in the range $1 \times 10^{-1} - 1 \times 10^{-5}$.

A training loop function iterates over the `DataLoader`. For each batch:

- 1) the cached encoder features and normalized labels are retrieved;
- 2) the head predicts a four-dimensional vector;
- 3) MSE loss is computed between predictions and labels;
- 4) gradients are back propagated, and the optimizer updates the head parameters (in training mode).

If available, mixed-precision training with `torch.cuda.amp.autocast` and gradient scaling is used on GPU to further reduce memory usage. After each epoch, the model is evaluated on the validation set and training and validation MSE values are recorded.

F. Emotion Attribute Prediction Pipeline

For inference, a helper function loads a single audio file, preprocesses it, runs it through the same feature extractor, frozen encoder, and trained head, and returns a dictionary of predicted normalized emotion scores. These are then denormalized using the known attribute ranges for:

- `Valence_best`,
- `Arousal_best`,
- `Submissive_vs._Dominant_best`,
- `Serious_vs._Humorous_best`.

The final outputs correspond to continuous estimates in the original attribute scales, making them suitable for interpretation and downstream analysis.

V. EVALUATION APPROACH

The model is evaluated using both global metrics across all four attributes and per-attribute metrics computed on the validation set:

- **Mean Absolute Error (MAE):** average absolute difference between true and predicted values, giving an intuitive sense of typical error magnitude;
- **Mean Squared Error (MSE):** average squared difference between true and predicted values, emphasizing larger deviations and reflecting overall stability;
- **Concordance Correlation Coefficient (CCC):** combination of correlation and agreement between predictions and ground truth, computed per attribute to assess how well the model follows the pattern and scale of each emotional dimension.

An evaluation function aggregates predictions and targets over the validation set, computes MAE, MSE, and CCC for

each attribute, and prints the results. These metrics together indicate whether the model predicts the correct intensity (low MAE and MSE) and follows the correct trend and scale (high CCC) for valence, arousal, dominance, and serious–humorous dimensions.

VI. OUTPUT (RESULTS)

The final model configuration employs a frozen Wav2Vec2–base encoder with a trainable GRU-based regression head and no sigmoid activation at the output layer. The model was trained for 14 epochs, after which stable convergence was observed.

The dataset consists of a total of **6,389** audio–annotation pairs, of which **5,751** samples were used for training and **638** samples were reserved for validation. This split ensures that model performance is evaluated on unseen data while maintaining sufficient training coverage.

At the end of training, the model achieved low validation error across all emotional attributes. The model with the best results finished training with an MSE on the validation set below 0.46, which was an error rate the model would not achieve when training with poorly set hyper parameters. The validation performance is summarized in Table I. These values represent averages across the four predicted attributes: Valence, Arousal, Submissive vs. Dominant, and Serious vs. Humorous. The labels that are being predicted were all normalized to be between 0 and 1.

TABLE I
BEST MODEL PERFORMANCE (NO SIGMOID ACTIVATION)

Attribute	MAE	MSE	CCC
Valence	0.0433	0.0052	0.3577
Arousal	0.0570	0.0062	0.7138
Submissive vs. Dominant	0.0331	0.0017	0.6086
Serious vs. Humorous	0.0572	0.0055	0.6513
Macro Avg.	0.0477	0.0047	0.5829

To illustrate training dynamics, Table II reports validation MSE at selected epochs during training. Validation error decreases rapidly in early epochs and gradually stabilizes toward the end of training, indicating smooth convergence of the regression head without signs of overfitting.

TABLE II
VALIDATION MSE PROGRESSION ACROSS TRAINING (FINAL CONFIGURATION)

Epoch	Validation MSE
5	0.0070
10	0.0061
14	0.0056

In addition to error-based metrics, distribution-level statistics were analyzed. The predicted mean values closely matched the validation means, indicating limited prediction bias. However, the predicted standard deviations were consistently smaller than those of the ground-truth labels, suggesting

conservative predictions centered around average emotional intensity.

Metric	Valence	Arousal	Dominance	Humor
Validation mean	0.5247	0.3107	0.5295	0.2017
Predicted mean	0.5185	0.3211	0.5334	0.1963
Validation std	0.0778	0.1108	0.0479	0.0836
Predicted std	0.0503	0.0834	0.0302	0.0688

TABLE III
VALIDATION AND PREDICTED STATISTICS FOR EACH DIMENSION

Overall, the low validation error and stable training behavior indicate that the final model effectively learns mappings between speech representations and continuous emotional attributes.

VII. ANALYSIS

The experimental results demonstrate that a frozen self-supervised speech encoder can effectively support continuous emotion regression when paired with a lightweight temporal regression head. When the model was trained for many hundred epochs, overfitting was evident. The validation MSE decrease, but then it would slightly increase and plateau between 0.005 and 0.006 while the training MSE would become several times smaller (without significant regularization). When training was stopped at a moment with a small validation error the resulting model much higher accuracy. Different values for weight decay were tested to reduce overfitting and 0.07 had the best results, even though overfitting would still occur. Additionally, the sigmoid activation function for the final model outputs turned out to cause worse performance. When the sigmoid activation was removed, leaving no final activation function, the performance improved. This likely happened because the sigmoid is not very good at predicting a distribution with a small standard deviation. Sigmoid activation is better for classification when values need to become very close to 0 or 1.

An analysis of the true and predicted statistics reveals that the predicted mean values closely match those of the validation labels. However, the predicted standard deviations are consistently lower than those observed in the ground truth. This reduced variance suggests that the model produces conservative predictions clustered around average emotional intensities, which is a common characteristic of regression models trained with frozen encoders and limited head capacity. In general, the model shows a good understanding of the spread of the emotions’ distributions.

a) Attribute-wise Performance: Performance varies across emotional dimensions. Arousal and humor achieve relatively stronger performance, reflected by lower error and higher CCC values. These attributes are likely closely linked to prosodic cues such as pitch, energy, and temporal dynamics, which are well captured by Wav2Vec2 representations. In contrast, Valence may be more subtle and context-dependent, making it harder to infer from acoustic information alone resulting in significantly lower correlation scores.

b) *Analysis of CCC Performance:* Although the model achieves low MAE and MSE across all emotional attributes, the Concordance Correlation Coefficient (CCC) values are comparatively lower, for Valence and Dominance. This behavior can be partially attributed to the reduced variance observed in the predicted outputs, as the frozen Wav2Vec2 encoder and lightweight regression head tend to produce conservative estimates centered around the mean. CCC jointly measures correlation and agreement in scale, and therefore penalizes predictions that underestimate emotional extremes, even when absolute error remains low. However, this does not fully explain why valence has a much lower CCC since its standard deviation is bigger than it is for dominance and it is close to the standard deviation for humor. Higher CCC values for Arousal and Humorous indicate that these attributes are more strongly encoded in prosodic and temporal speech features, which are effectively captured by the pretrained representations.

Overall, the analysis highlights a trade-off between numerical accuracy and dynamic range modeling in resource-efficient architectures. While freezing the encoder enables efficient training and stable convergence, it also limits the model’s ability to fully capture extreme emotional variations, which is reflected in CCC-based evaluation metrics.

The absence of sigmoid activation avoids saturation effects and allows the model to represent continuous emotional intensity more flexibly, contributing to smoother optimization and improved numerical stability.

VIII. CONCLUSION

This work presents an efficient approach for continuous audio emotion recognition using a frozen Wav2Vec2-base encoder and a lightweight GRU-based regression head. By removing sigmoid activation from the output layer, the model achieves stable training behavior and low validation error while maintaining minimal computational overhead.

The final results demonstrate that self-supervised speech representations encode sufficient emotional information to support multi-attribute regression without full model fine-tuning. Although the model exhibits conservative prediction behavior, it achieves reliable performance across multiple emotional dimensions.

Future work may explore partial encoder fine-tuning, more expressive temporal modeling, or multimodal extensions incorporating textual or visual information to improve sensitivity to extreme emotional variations.

IX. FUTURE WORK

Several directions can be explored to further improve the proposed emotion recognition system. First, partial or full fine-tuning of the Wav2Vec2 encoder may help increase the dynamic range of predictions and improve CCC, particularly for subtle attributes such as Valence and Dominance. Selective unfreezing of higher transformer layers could provide a balance between performance gains and computational efficiency.

Second, more expressive temporal modeling techniques, such as transformer-based pooling or multi-head attention

mechanisms, may better capture long-range emotional dynamics compared to the current GRU-based head. Additionally, incorporating variance-aware loss functions or regularization strategies may help reduce conservative prediction behavior and improve sensitivity to extreme emotional values.

Finally, the approach can be extended to multilingual emotion recognition by training and evaluating the model on speech from multiple languages. While emotional prosody shares universal characteristics, linguistic and cultural differences influence emotional expression and perception. Leveraging multilingual pretrained speech models or language-adaptive fine-tuning strategies could improve robustness and enable cross-lingual generalization. This direction would also allow investigation into whether a single unified model can generalize across languages or whether language-specific calibration is required.

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